TECHNICAL SPECIFICATION OF MPLS-TP FOTE & NMS

1. SCOPE OF SPECIFICATION

This specification is to cover planning, design, engineering, supply, installation, termination, testing, commissioning of Fiber Optic End Terminal Equipment with MPLS-TP Technology **& integration with NMS**. **The system shall provide one single platform for a whole range of applications from 1G bits/sec to 10G bits/sec Ethernet traffic under MPLS-TP technology. The Contractor shall offer type tested equipment as per requirement of technical specifications.**

(i) Integrating the new F.O. links with the existing (SDH&PDH) F.O. network.

(ii) MPLS-TP Compatible Equipment along with suitable interfaces etc.

(iii) Provision of an efficient Network Management System licensed to accommodate at least 100 Network Elements (MPLS-TP) and upgradable to accommodate at least 300 Network Elements. All cabling, wiring, patch facilities, equipment DDF's and interconnections to the supplied equipment side of the defined interfaces.

(iv) Integration of MPLS-TP equipment with Network Management System (NMS) and other make SDH FOTEs.

(v) VOIP system integrating with the Hybrid network using Ethernet Ports of MPLS-TP Equipment, existing EPAXs, proposed EPAXs with Dual Server architecture at Vidyut Soudha, Vijayawada.

(vi) All cabling, wiring, Digital Distribution Frame patch facilities, equipment MDF and interconnections to the supplied equipment at the defined interfaces.

(vii) MDF &DDF cross connects required to route and activate circuits.

(viii) System integration of the supplied sub-systems.

(ix) Supply of mandatory spares.

(x) All other associated works and items described in the technical specifications for a viable and fully functional communication network.

(xi) The system shall offer complete solution for SCADA on IEC 60870-5-104, Voice using VOIP, Video conferencing, Phasor Measurement Unit (PMU) (Future), differential and distance tele-protection as per IEEE 37.118 standard, and RTU/SAS data, including IP/Ethernet and status/control signals.

(xii) Supply, laying, terminations along with cable trays, tags, and connectors of all DC/AC power cables, control cables, communication, F.O. patch chords & Earthing cables necessary for commissioning the FOTE equipment and NMS as per site requirement for a fully functional system in this tender is in the scope of bidder. For AC supply: 3x2.5sqmm Multistrand Copper unarmoured cable. The minimum requirement of cable size for DC supply is 2x4 sq.mm multi strand copper unarmoured cable (for each DC source) approximately 25mtrs of Power cable will be required for each Substation.

(xiii) Power supply to all the equipment supplied under this contract is in the scope of the Bidder.

2.0. GENERAL REQUIREMENTS:

Introduction:

The proposed equipment shall be designed to operate in electrical high-voltage networks and suitable for installations in substations with harsh environments and high electro-magnetic interference. It shall be highly reliable and provide secure communication. The MPLS-TP equipment shall be designed and field proven to ensure maintenance free operation and comply with the environmental conditions in the substation environment.

The equipment offered shall comply with the latest ITU-T and IETF recommendations, ETSI and IEEE standards. The equipment shall comply with the environmental conditions in the substation environment. It shall also with stand extreme temperature conditions and a particle environment. The equipment shall have a very high MTBF.

On PACKET TRANSPORT LEVEL, interfaces for optical L2/MPLS-TP based transmission minimum 1Gbit/s shall be available. Enhanced traffic engineering using L2/MPLS-TP technology as per relevant IETF standards shall be supported by providing VPWS, VPLS services using the L2/MPLS-TP infrastructure. The equipment shall support high end security and industry standard encryption features with lower latency and higher through put.

End to end channel supervision should be supported. The equipment shall have a combination of MPLS-TP and L2 packet switched technology with traditional legacy interfaces and shall be complemented with synchronization and cyber security features.

The platform shall have the means to cross- connect legacy services into one or several bundled E1 streams for circuit emulation over packet switched network/SAT operations as and when required by addition of modules.

Enhanced traffic engineering using MPLS-TP technology as per relevant IETF standards and comply with all service standards and shall support E-Line, E-LAN, E-Tree and E-Access services.

Static configuration of bidirectional and co-routed communication channels shall be supported using the NMS. Dynamic routing shall be prohibited for all critical data channels. End to end channel supervision should be supported. The equipment shall be software controlled, of modular designand all modules shall form an integrated part of a 19” shelf. A separate rack arrangement shall be provided for housing the MPLS-TP equipment as per the standard adopted for SDH F.O. end terminal equipment. Each network element shall be manageable locally as well as from an operation center.

EQUIPMENT TECHNICAL REQUIREMENTS:

Description:

The proposed MPLS-TP fibre optic communication equipment should provide a Network with aggregation layer at MPLS-TP traffic along with legacy TDM interfaces and a core /Backbone Network on MPLS-TP technology.

The primary function of the communication network is to provide a highly reliable voice, video conferencing, tele-protection and data communication system in support of the SCADA/EMS. The communications support requirement for SCADA/EMS systems is for low & high-speed data, express voice circuits and administrative voice circuits & VOIPs.

The communication system shall provide data & voice(VOIP) connectivity across the various locations and connectivity of RTUs with Sub- LDCs and Sub-LDCs with MLDC.

A brief summary of the communication system requirements is as follows:

(a) Standalone distance tele-protection commands interface preferably fully manageable from a common control center.
(b) Data transport supporting Network Management channels in MPLS-TP technology.
(c) The connectivity envisaged between RTUs/SAS and Control Centre in Wide Area Network(WAN) on TCP-IP using IEC610870-5-104 protocol. The equipment shall support digital VOIP subscriber & gateway functionality (SIP-Session Initiation Protocol) by connecting an integrated/external VOIP gateway.
(d) Integrated differential protection functionality fully manageable under the same NMS.

Management: Remote management system with IP/Ethernet interface.

Redundancy Capabilities:

The equipment shall provide enhanced redundancy features in order to ensure highest availability of the communication network. The following redundancy schemes shall be supported (Hot standby) as per the BoQ:

- Redundant power supply unit or Distributed Power Supply
- Enhanced PSN traffic protection schemes
- Minimum1:1LSP protection switching

Telephony:

To support Digital VoIP subscriber and gateway functionality (SIP) by Connecting an external VoIP Gateway.

Traffic Protection Scheme:

1:1 Path protection

Minimum1:1 protection of MPLS LSPs shall be supported. Switching from main to backup path shall be automatic, bidirectional and symmetrical. Configuration options shall exist for reversible or non-reversible operation. The main requirement of the network protection mechanisms is that they should be able to recover traffic very quickly (switching time should be less than 50msec as per MPLS-TP standard). Further, protection should also operate autonomously from the network operation center (without involvement of control plane or management system).

Synchronization:

- It shall be possible to synchronize the equipment using an external clock source, derived from a network or from an internal oscillator. The synchronization shall be configurable and it shall be possible to distribute the synchronization to other equipment as well. The system shall have the means of switching to select the synchronization source as well as a means of preventing the system from creating synchronization loops.
- The equipment shall support synchronous Ethernet and a synchronization selection process based on ESMC (Ethernet Synchronization Message Channel).
- The precision time protocol (PTP) as defined by IEEE/IEC shall be supported for frequency and time of day synchronization.
- The equipment shall support PTP transparent clock and PTP boundary clock functionality. The equipment shall be able to be synchronized to a GPS grandmaster and provide time of day information to other devices by using IRIG-B or SNTP.
- GPS Clock with NTP server should be supplied with NMS
- Configuration support for Sync E and PTP synchronization shall be available **e.g. by means of wizards or guided configuration.**
- Synchronization-Multiple type soft timing references should be available from multiple timing sources:
- E1/T1/IECC.37.94 interfaces.
- An external input of 2.048MHz
- Synchronous Ethernet (SyncE) from GBEor10GE.
- G.813 internal clock,free-run,accuracyof±4.6ppm
- Precision Time Protocol(PTP)-IEEE1588v2PTP
- Used in multicast Ethernet encapsulation modeITU-TG.8275.1
- Must support atleast the following1588v2 PTP types
- Boundary Clock(BC)
- Ordinary Clock(OC) Master
- Ordinary Clock(OC) Slave
- Transparent Clock (TC)
- Synchronization Status Message(SSM)support
- Jitter and Wander according to ITUTG.823, G.825.
- The system time must support PTPv2 as the Time-of-Day (TOD) source (RFC 6374) for OAM time stamping.

The following GBE interfaces as per the distance given in BoQ should be supported:

- 1000 Base X
- SX, LX, EX, ZX, Bi-directional, preferably coloured.
- 10/100/1000BaseT/preferably GPONONU interface

The following SFP+1 GE interfaces should be supported:

- ER ,ZR ,SR ,LR ,BD ,preferably Colored &Tunable

All SFPs and SFP+ shall be of MSA 2000 compliant in order to work with any vendor FOTE equipment that is of MSA 2000 compliant.

The offered equipment should support the following environmental conditions

a. operation mode:

- ETSI-300-019-1-3Class3.1
- Temperature range:-5°Cto+55°C
- Relative humidity:5%to95%
- The offered equipment should support the following:
- EMC according to EN300386/2000,IEC61850-3,IEC61000-6-2
- Safety according to EN60950/2000
- Laser Safety according to IEC60825-1&2(AS/NZS2211.2)

b) In particular the following type tests should be complied as per IEC 61850-3/ equivalent IEC /EN Standard .

a. Power

Input: Auxiliary Voltage

b. Insulation Tests:

i. Dielectric Tests

ii. Impulse Voltage with stand Test

C. Influencing Quantities

i. Limits of operation

ii. Permissible ripples

iii. Interruption of input voltage

e. Electro magnetic Compatibility Test:

i. 1MHZ.burst disturbance test

ii. Electrostatic Discharge Test

iii. Radiated Electromagnetic Field Disturbance Test

iv. Electrical Fast transient Disturbance Test

v. Conducted Disturbances Tests induced by Radio Frequency Field

vi. Magnetic Field Test

vii. Emission (Radio interference level)Test.

viii. Conducted Interference Test

f. Functional Tests as applicable for MPLS–TP equipment

g. Environmental tests:

i.Cold Temperature

- ii. Dry Heat
- iii. Wet heat
- iv. Humidity(Damp heat Cycle)
- v. Vibration
- vi. Bump
- vii. Shock

Cyber Security:

The equipment shall support integration into a cyber-security architecture enabling cyber secure operation of communication network. The equipment shall be type tested for cyber security robustness by DSAC, a Device Security Assurance Center/ Cyber Security Certification obtained from Indian Govt. approved organizations. Type test certification by DSAC, a Device Security Assurance Center/ Cyber Security Certification obtained from Indian Govt. approved organizations, for cyber security robustness shall be done.

- Authentication ,Authorization and Accounting(AAA)
    - Local management port disabling

- User roles (privilege-levels)
- Radius based authentication
- Security relevant local event logging (audit log)
- Sys log

- Secured management communication
  - SNMPv3
  - SSH
- Denial of service (DoS) prevention for device functions and user traffic
  - Rate limiters for different traffic types
  - Firewall towards device management
- Traffic authentication.
- Access Control Lists(ACL)

The equipment shall be of robust design and hardening shall be done considering cyber security aspects.

- Preferably 802.1xdynamic port-based user authentication
- MAC level Port security
- In P2MPservice, spokes must not be able to talk to each other, unless other-wise configured.
- In MP2MP service, it must be possible to limit the number of MAC states a customer can inject into Edge Devices on a per VPN per end-point basis.
- OSPF Encryption must be supported for the equipment management.

Cyber Security Compliance:

The fibre optic terminal equipments proposed in this tender shall comply all the cyber security requirements defined in the IEC standard 62443-3-3.Necessary type test report obtained from Indian Govt. laboratory (CPRI)/ Authorized International laboratories for the above cyber security features shall be uploaded with the bid.

2.2.7.2 Other mandatory test:

As per the order dated 02.07.2020 of the Ministry of Power, Govt of India that apply to any item imported for end user or to be used as a component or as a part in manufacturing, assembling of any equipment or to be used in power supply system or any other activity directly or indirectly related to power system the compliance to following requirements are to be ensured. In this context all expenses towards the mandatory testing is to be borne by the project contractor.

d) All equipments, components and parts imported for use in the power supply system and network shall be tested in India to check for any kind of embedded malware/ Trojans/ cyber threat and for adherence to Indian standards.

ii) All such tests shall be done in certified laboratories that will be approved by the ministry of power, Govt of India.

iii) Any import /equipment /components /parts from:"Prior reference" countries as specified or by persons owned by controlled by or subject to the jurisdiction or the direction of these "Prior reference" countries will require prior permission of the Govt of India.

iv) Where the equipment / components /parts are imported from "Prior reference"

countries with special permission, the protocol for testing in certified and designated laboratories shall be approved by the by ministry of power, Govt of India.

It is mandatory for the bidders/OEM from prior reference countries to obtain the type test report for having tested for cyber security features as per relevant standard from CPRI laboratory of India at their own expenses. As per CEA guidelines 2021, "All ICT based equipment/system deployed in infrastructure/system mandatorily CII are sourced from the list of the "Trusted Sources" as and when drawn by MoP/CEA".

2.2.8.MTCTE Certification:

As per recent MoP guidelines MTCTE certification has to be obtained by OEMs /Importers before sale of Notified Telecom, Networking products and the same has to be furnished by the successful bidder before the award of work. The bidder shall furnish a undertaking in the offer towards submission of MTCTE certification before the award of work.

2.2.9 Quality of Service

Hardware support to ensure device internal and user traffic quality of service shall be provided.

- ≥8 hardware queue per physical port
- Configurable scheduling profiles
- Priority mapping tables (DSCP, PCP, EXP, PHB, TC,Source/Destination MAC, etc.,)
- Priority configuration for individual protocols (e.g.PSC,MCC,…)
- Port and service based rate limiting (atleast single rate two color marker (SR2CM))
- Configurable to remote control for unicast, unknown unicast, multicastand broadcast per port.
- QoS (IEEE802.1p)VLAN services including QinQ (IEEE802.1ad)

Aggregation/Transport InterfaceRequirement.

MPLS-TP Functionality.

The following MPLS-TP functionality shall be supported:

- The equipment shall be capable of operating as a label edge router (LER) and label switch router (LSR) in MPLS-TP networks.
- Static configuration of MPLS-TP tunnels via configuration tool as well as network management system
- Bi directional MPLS-TP tunnels
- Priority handling usingL2/L3 class of service information
- Mission critical services &guaranteed performance parameters for tele- protection applications meeting IEC60834-1 standard for distance protection along with security, dependability and transmission time using **stand-alone protection coupler connected to MPLS-TP Devices on Ethernet/E1 Ports.**
- Mission critical services & guaranteed performance parameters for differential

Tele-protection applications meeting C 37.94 standard. For differential protection, relay to relay communication channel shall be provided through MPLS-TP equipment using IEEE C37.94 optical interface.

- The following topologies shall be supported.
- Point to point virtual private wire services (VPWS)
    - Tree structures
    - Virtual private LAN structures (VPLS)
    - Hierarchical VPLS structures (H-VPLS)
- End to end Operation Administration and Maintenance (OAM) channel with at least the following functionality
    - Continuity check messages (CCM)
    - Remote defect indication (RDI)
    - Route tracing (LSPping /trace route)
- 1:1traffic protection with<50ms switch over time
- Management functionality using dedicated MPLS channel.

- Port based MPLS assignment
- VLAN based MPLS assignment
- Circuit emulation of legacy services via MPLS-TP shall be supported..
- Monitoring through put function facility for all channels to know the load & traffic up to the lower level of the network.
- Mission critical services & Guaranteed performance parameters for tele- protection applications meeting IEC 60834-1 standard for distance protection along with security, dependability and transmission time using standalone protection coupler connected to MPLS-TP Devices on Ethernet Ports.

Ethernet Requirements:

- The proposed network shall support 802.3-2005 frame format
- The proposed network shall support automatic MAC address learning without impact on line-rate performance.
- The proposed equipment shall have Minimum6x1Gbit/s optical ports &2 x 1 10Gbit/s Optical Ports as per requirement shall be available for Ethernet uplink purposes.
- All 1 Gbit/s /10Gbits/s optical shall be available for Ethernet uplink purposes as well. The uplink 1Gbit/s/10Gbits ports must also support MPLS-TP encapsulation.
- Ethernet electrical connection:10/100/1000BaseT
- Ethernet optical connection:1000Base-X
- Protection by means of xSTP
- Port and VLAN based rate limiting
- VLAN based forwarding

- Supports Classification of Service (CoS)prioritization with in a singleVLAN.
- Supports QoS, optimizing Bandwidth allocation and simplifying network infrastructure.
- Priority handling(PCP/TC/PHB/EXP)shallbeconfigurable
- Port based access(all traffic is mapped into MPLS-TPservices)
- VLAN based access(separation o VLANs into different MPLS-TPservices)
- Mix of port and VLAN based access and mapping into VPWS on the same physical port
- Traffic type filtering shall be supported (multicast, broadcast,unicast)
- VLAN mapping shall be possible
- SNMP MIBs.
- The proposed network shall support 802.3-2005 frame format
- The solution must support 1Gbps Ethernet client interfaces using pluggable optics.
- The proposed network shall support disabling MAC learning for point to point (E Line) Ethernet virtual circuits
- The proposed network shall support static MAC FIB entries as part of policy forwarding.
- The proposed network shall support a jumbo Ethernet frame of more than 9000 bytes; Supplier to mention the max size supported.
- The solution must support EPL, EVPL, E-LAN and E-TREE services.
- The proposed network shall support the participation and discard options for IEEE 802.3x Flow Control on access ports. Policing and bandwidth allocation.
- Preferably the proposed network shall support 802.1x authentication on UNI Ethernet ports.
- The proposed network shall support tunnelling and discard options for all Bridge management groups of protocols.
- Preferably the proposed network shall support programming Ethernet type value for Q-in-Q tunnelling globally and on port-basis.
- Preferably the proposed network shall support VLAN based VLAN stacking/ tunnelling with the ability to support multiple outer VLAN tunnels per E-UNI. The proposed network shall be able to apply multiple outer VLAN tags on a E-UNI based on inner/customer VLAN ID if the customer frame is tagged or a configurable VLAN ID for untagged frames.
- The proposed network shall support VLAN ID Translation (not tunneling/stacking), i.e., translating VLAN ID received from one node to a different (network) VLAN ID at originating E-UNI and translating back to customer VLAN ID at terminating E-UNI.
- Ethernet circuits must be completely transparent to any traffic. Detail any traffic types that are not transparently carried across circuits.

- The solution should support color-aware and color-blind policing.
- The solution should support Broadcast Storm Control (BSC)
- Protected services must have adversely routed back up path to which traffic is switched to in less than 50ms when the primary path fails.
- Protected services must failover to the backup path based on complete failure of the primary path and based on partial traffic loss (signal degradation).
- The solution must support arbitrary network topologies including rings, mesh and partial mesh topologies.
- The solution must support VLAN swapping for EVPL services such that each side of an EVPL may use different VLANs or one side may be untagged with the other being tagged.
- The solution shall support data link OAM:ITU-TY.1731or equivalent IETF G-8113.1 for Fault management and performance monitoring and covering Fault Management, Proactive Connectivity , Verification , Continuity Check, and Remote Defect Indication for MPLS-TP Equipment.
- The solution shall support 802.3ah EFM.
- The solution must support transparent transport of customer class of service markings when present, including 802.1p, IP DSCP and MPLS EXP
- Preferably the proposed network shall have UNI (User Network Interface) and NNI (Network Node Interface). Typically, UNI would be configured in 802.1Q mode or Q-in-Q mode and NNI shall be in MPLS-TP(P2P)or VPLS/H-VPLS (MP2MP) mode. In order to simplify the operation of the product tis intended that no restriction be placed on UNI and NNI ports and all the ports shall be capable of supporting any mode of operation.
- The proposed network shall support configuring all 4094VID on all ports and at the same time support all 4094 VLANs simultaneously. The operator must be able to reuse the same VLAN-ID on a different port on the same switch and terminate it into a different PW.
- The proposed network shall support the capability of designating an Ethernet port as a primary and another as a backup for that port. When the primary port is active, The proposed network will always receive and send traffic on the primary port. The proposed network could receive traffic on the backup port, but will not pickup that traffic as long as the primary port is active. When the primary port fails, the backup port becomes active and The proposed network immediately starts receiving and transmitting traffic on that port. The application of this requirement is to allow 1:1 redundancy on the client side. The proposed network shall support the primary and back up ports being on different slots. When the primary port recovers after a failure, The proposed network shall revert the primary port to the active state and the backup port to the standby state.

Ethernet over MPLS-TP Requirements

- The proposed network shall support Ethernet MPLS Pseudo Wires per RFC 5462.

- The proposed network shall support Ethernet MPLSPWs for Ethernet frames arriving over physical Ethernet interfaces.
- The proposed network shall support the configuration of IEEE 802.1p user priority bits to MPLS PW EXP bit mapping on ingress when performing MPLS encapsulation. The proposed network shall be able to write the EXP bits accordingly.
- The proposed network shall support static setup of Ethernet MPLSPWs
- The proposed network shall transparently pass customer BPDU control frames over MPLSPWs, by configuration on an attachment circuit level (e.g., VLAN ID, Ethernet port)
- Preferably the proposed network shall support mapping of VLAN Tunneled/Stacked (Q-in-Q) Ethernet frames to a pseudo wire based on double look-up of outer and inner IEEE 802.1q VLAN IDs.
- The proposed network shall support the configuration of IEEE 802.1d user priority bits + DEI to MPLS PW EXP bit mapping when performing MPLS encapsulation. The proposed network shall be able to write the EXP bits based on the outcome of this mapping.
- The proposed network shall support MAC address flushing across the VPLS instance when it learns of a local topology change on the access network.
- MPLS-TP should allow pseudo-wire services for the following Ethernet Services:
- Ethernet Line (E Line) for P2P connectivity, used to create Ethernet private line services, Ethernet based internet access services, and P2P Ethernet VPNs. These include:
- Ethernet Virtual Private Line (EVPL):P2P connectivity over shared bandwidth .Service can be multiplexed at the user to network interface (UNI) level.
- Ethernet LAN (E LAN) for multipoint to multipoint (MP2MP) (any to any) connectivity, designed for multipoint Ethernet VPNs and native Ethernet Transparent LAN Services (TLS). These include:
- Ethernet Private LAN (EPLAN): Multipoint connectivity over dedicated bandwidth, where each subscriber site is connected to multiple sites using dedicated resources (so different customers' Ethernet frames are not multiplexed together).
- Ethernet Virtual Private LAN (EVPLAN): Multipoint connectivity over shared bandwidth, where each subscriber site is connected to multiple sites using shared resources. This is a highly cost effective service, as it can leverage shared transmission bandwidth in the network.
- Ethernet Virtual Private Tree (EVP Tree): An EVP Tree is an E Tree service that provides rooted multipoint connectivity across a shared infrastructure supportingstatisticalmultiplexingandoversubscription.EVPTreeisused for hub and spoke architectures in which multiple remote offices require access to a single headquarters, or multiple customers require access to an internet SP's point of presence (POP).

- The proposed network shall support rate limiting (policing)on all Ethernet interfaces per RFC 2697 and 2698.
- The proposed solution shall provide traffic management support with the following key functionalities:

- Dedicated traffic manager role in NMS enabling definition of traffic management related parameters for services.
- Service profiles, enabling a traffic manager to create “templates”for users configuring services
- Queueing profiles and traffic scheduling algorithm should be configurable
- QoS with ratelimiting and optionally hierarchical QoS

- The proposed network shall support L2 Ethernet classification based on physical port, IEEE 802.1q service VLAN ID (EVC), 802.1p user priority bits field value (EVC plus CoS), and IP DSCP/precedence bits (EVC plus DSCP). Any of these fields could be a wildcard in a given configuration.
- The proposed network shall support VPLS (Virtual Private LAN Service)
- The proposed network shall support Hierarchical VPLS(H-VPLS)
- The proposed network shall support Q-in-Q as access to a virtual switching instance
- The proposed network shall support Ethernet MPLS PWs as access to a virtual switching instance
- The proposed network shall support multi-class PWs with mapping between 802.1p bits and MPLS PW exp bits as defined for Ethernet MPLS PWs
- The MPLS Packet Transport shall support mapping between incoming exp bits on MPLS PW spokes to VPLS MPLS PW exp bits on ingress upon transmission over the VPLS and on egress upon transmission on the spokes

MPLS-TP Requirements

- The proposed network shall support Bidirectional Forwarding Detection (BFD)for MPLS LSP as defined in IETF draft draft-ietf-bfd-mpls-04.txt or any subsequent revision of that draft
- Setting BFD message frequency for both the transmission and minimum receive interval shall be as low as 3.3, 10, 100 m secs.
- MPLS-TP implementation should be based on NMS based provisioning of services and LCT based provisioning of services.
- Tunnel OAM should be based on BFD.
- The proposed network shall support configuring the exp field value of the MPLS BFD messages
- The proposed network shall support MPLS ping in both ping and trace mode as specified in IETF RFC 6426
- The proposed network shall support egress shaping per LSP.
- The proposed network shall support user configurable mapping between outer MPLS exp bits and Ethernet user priority bits per 802.1d

- The proposed network shall support MPLSPW encapsulation over GFP-F
- MPLS-TP OAM will be based on the GAL/G-ACH concept) and must work independently of the control plane.
- OAM must work within the data plane, forward and backward traffic must follow the same path and optimal protection mechanism for all topologies including ring and star topologies (Segment, Ring and Linear protection).
- BFD OAM should be according with theITU-TG8113.2or later relevant document (IETF)
- The solution should use Hierarchical methods to reduce and support the mesh of tunnels / Paths.
- The proposed network shall support static MPLS label addition, swapping and popping as specified in RFC 3032.
- It shall support configuring failure detection time in absolute time or as a multiple of the receive frequency
- It shall support Proactive Connectivity Verification, Continuity Check ,and Remote Defect Indication for the MPLS Transport Profile as per RFC 6428
- The proposed network shall implement class based queuing at each logical interface.
- Preferably the proposed network shall support hierarchical Queuing whereby a diffserv scheduling system can apply on an LSP basis.
- Preferably the proposed network shall support the definition of a Diffserv scheduling system per logical interface.
- The proposed network shall support rate limiting per interface and per sub interface on all interfaces.
- The proposed network shall provide the capability of setting the MTU size
- The proposed network shall provide the capability for configuring the PWID
- The proposed network shall support binding a PW to an LSP.

OAM Requirements

- The OAM support should be based on recommendations in G.8113.1 and G.8113.2. Operations, administration and maintenance mechanisms for MPLS-TP networks using the tools defined for MPLS.
- Preferably the system shall support Ethernet OAM in full compliance with 802.1ag, 802.3ah , Y.1731 or G 8113.1 covering Fault Management, Proactive Connectivity , Verification , Continuity Check, and Remote Defect Indication.
- The proposed network shall support a full suite of OAM features and functionality to easily monitor, administer and troubleshoot all the services.
- The proposed network shall support OAM framework as defined in RFC 6371 and requirements defined in RFC 5860
- The system shall support OAM for

a) Detection & recovery of Path Defects

b) Diagnosis of a Broken Path

c) Preferably Service Level Agreement Measurement forJitter,Packet Loss and Latency.

d) Frequency of OAM Execution

e) Inter-working for Fault Notification

f) Detection of Denial of Service Attacks

g) LSP/PW ping & traceroute

h) Preferably VPN ping.LSPPingfor OAM as per RFC6426 standard through MPLS-TP tunnels

i) Preferably SNMP traps.

j) Mac Pinging – Continuity Check LSP Ping for OAM as per RFC6426 standard through MPLS-TP tunnels. Continuity check or Bi-Directional forward detection must be included in MPLS-TP Tunnels

k) Trace-route

l) Loop-back Connectivity check

m) Ethernet–Local Management Interface(E-LMI)

n) Maintenance End Points(MEP)on switchports

o) Maintenance Intermediate Points(MIP)

p) ConnectivityFaultManagementover802.1adlinks

q) SNMP traps

r) Preferably the System shall support Link Layer OAM (802.3ah) with following functionalities or as per relevant IETF standards

OAM discoveries
Link Monitoring
Remote Loopback
Remote Fault Indication

- The OAM packet should interoperate with third party network element All attributes of OAM packets should be understandable by third part network elements(attributes like timers, no.of retries, dead interval, enable, disable OAM etc)
- OAM should inform the management system / third part application about the status of service, connection per box basis and per user basis. Details information should be available like ingress and egress value, connection ID, port ID.
- There should be mechanisms to fetch delay, jitter and round trip delay, packet loss and frame loss for MOS, unicast & multicast.

- There shall be a mechanism to measure delay, jitter and round trip delay, packet loss and frame loss end-to-end for a given flow by any means. The system should synchronize the counters on both ends.
- The exchange of OAM packets should be end-to-end with provision to count the number of times connections were down. It should identify the status of the link.
- OAM should provide a mechanism for trace-route to help operators to trace the path from ingress to egress network element with details of intermediate element.
- The OAM packet should be isolated to the user's datapacket and it should be given highest priority.
- OAM packet should have attributes pertaining to ingress and egress filter.
- OAM should run per connection basis. The connection can be made using VLAN, Port, or other attributes.
- The solution shall support debugging through Event Logs, Alarm Logs, Performance logs, debug files in Equipment.
- Selective debugging of data plane aspects shall be supported including logging of individual ACL rules, FIB events, etc.
- Configuration management: backup/restore of configuration from a remote system using standard protocols like FTP,orTFTP. All of these functions shall be able to be performed through in-band interfaces and OOB management Ethernet port.
- SNMPv1, v2c and v3 shall be supported.
- Intensive OAM processing shall be supported on per LSP/PW basis(GAL/GAch for every LSP/PW),
- Sub-50ms PG (protection group) processing needs to be implemented on per LSP/PW basis.
- Continuity Check messages
- Remote defect Indication
- Route tracing
- The proposed network shall support802.1ag Connectivity Fault Management standard
- The proposed network shall discard customer frames that contain unassigned customer VLANs on 802.1q port (including untagged customer frames).
- The proposed network shall support counters for the number of broadcast, uni cast, multicast, and unknown unicast frames flooded per VLAN.
- The proposed network shall support MPLS-TP OAM requirements per RFC 6428
- The proposed network shall support the MPLS-TP OAM bundled with MPLS-TP tunnel for the monitoring and the protection.

- The MPLS OAM or BFD packet sending interval should be user configurable down to 3.3 ms.
- The proposed network shall support RMTF(Remote Fault)/FEFI(Far End Fault Indication) capability over management plane
- The proposed network shall support AIS and RDI (also 802.1ag)for ELine EVCs.
- The proposed network shall discard customer frames that contain unassigned customer VLANs on 802.1q port (including untagged customer frames).
- The proposed network shall support tunneling /discard of customer BPDUs

Scalability:

- Every UNI port must be able to support full range Customer VLAN IDs both in transparent and Channelized (mapped) mode. Different customers must be able to re-use the same range of CVIDs if they are attached to different UNIs.
- Every customer VLAN IDs (minimum 4089 CVIDs) on each physical port should have at least 8 queues for customer traffic /QoS handling.
- Each physical port should have buffering inbuilt. (Mention the amount of buffer per interface in terms of ms)
- The solution should use Hierarchical methods to reduce and support the mesh of tunnels / Paths.
- The solution should support multi segment pseudo wires.

Management Communication Channel:

- Support Management LAN interface for DCN.
- Support in-band MCC for Ethernet MPLS network.
- IPV4andOSPFV2should be supported.
- Management VLAN should be supported.

3.8.Security Requirements

- The proposed equipment shall support the following security features
- 802.1xdynamic port-based user authentication
- MAC level Port security
- SSH support
- Customer-facing ports(UNIs)on the CE device must be disabled and should not forward any traffic as default security behavior.
- Customer-facing ports (UNIs) on the CE device must not forward any traffic even after it is enabled, until it is attached to a service, e.g. PW
- Customer-facing ports (UNIs) must not allow access to Management Network, even though Customer may use/spoof combination of MAC addresses, VLAN IDs, IP Subnet (Address) that are legally allowed access to Management network.
- Customer-facing ports (UNIs) must ensure that they do not have access to other customer's traffic in any way.

- Network control Protocols must not be coupled with customer control protocols to maintain the stability of the Provider network.Customer control-plane traffic such as BPDU, ISL etc. must be tunneled end-to-end without interacting with SP control-plane.
- In P2MP service, spokes must not be able to talk to each other, unless otherwise configured.

Power Supply:

The equipment shall operate from a nominal 48V DC (-15% /+20%) battery. The equipment should be cut-off beyond this range and automatically restored in the operating range. The equipment /system shall support dual power feed or (1+1hardware protection)i.e.that two power sources can be connected directly to the equipment (two connection points) or dual power feeder. The operating voltage should be 48V DC.

The equipment shall supervise the power source and provide corresponding alarm in case of loss of either power source. Supervision relay having potential free contacts and audible alarm shall be provided. Necessary surge protection shall be provided to filter all surges.

4.0 Safety:

The equipment shall be safe to use and shall comply with EN 60950-1 and IEC 60255-27 Measuring relays and protection equipment product safety requirement.

5.0 Ambient Conditions:

Storage and transport:

- Temperature range: -25°C...+70°C
- Humidity: max. 95% (no condensation)

Operation;

- Temperature range: -5°C…+55°C
- Humidity: max.95%(no condensation)

6.0 Electro magnetic compatibility (EMC):

The equipment shall comply with the EN55022classA, EN61000-6-2, EN 61850-3, and shall be in conformance with CE.

7.0 Mechanical Construction

The equipment shall be available as a 19" shelf to be mounted in a 19" rack or19" cabinet. It shall be of robust design. All modules shall be integrated in the same shelf. All connectors shall be accessible from the front and comply with international specifications.

Equipment Sub Racks shall be steel/ aluminum fabricated and finished on all surfaces. All metal and welds shall be thoroughly cleaned and sanded to obtain a smooth finish .All surfaces shall be treated for rust and primed to form a bond between metal and the finish coats of paint. All equipment racks,enclosures and equipment, including distribution frames, shall be clearly labelled with unique identifiers consistent with Contractor supplied floor plans and rack elevations.

Dust and moisture protection shall be as per the IP41 specification, or better standard. The equipment cabinet shall be built in such a way that it prevents the entry of all type of rodents.

The Contractor is required to provide and install all additional necessary indoor and outdoor cable raceways. Signal cabling and power cabling shall require separate cable raceways. Signal and power cabling shall not share the same raceways and shall be installed as far apart as is practical. Adequate shielding shall be provided as required. Especially, all the F.O. patch cords from FODB to FOTE cabinet shall be enclosed with a minimum of 1" flexible standard steel/Aluminum hose. All the cables including power, control, communication cables etc.,will be in the scope of contractor/bidder as per the site requirement.

The Contractor shall be required to provide protection from lightning and transient voltages for all wideband communications equipment. Suitable surge voltage protection devices shall be installed in all panels. Proper design approval shall be obtained from APTRANSCO before supply.

All equipment provided under this specification, shall be physically mounted in sub-racks and cabinets (enclosures).The Contractor shall determine and propose for the Employer approval, the type, size, weight and manner of installation for each location.

Equipment Sub Racks shall be steel/aluminum fabricated and finished on all surfaces. All metal and welds shall be thoroughly cleaned and sanded to obtain a smooth finish. All surfaces shall be treated for rust and primed to form a bond between metal and the finish coats of paint. All equipment racks, enclosures and equipment, including distribution frames, shall be clearly labeled with unique identifiers consistent with Contractor supplied floor plans and rack elevations.

Dust and moisture protection shall be as per the IP41 IEC 60529 / IS12063 and the type tests and routine tests prescribed in IS/IEC 60947 Low - voltage Switchgear and Control gear standards shall be performed on the complete panel assembly.The degree of protection and the panel thickness shall be mentioned in the drawing. Typical panel drawing shall be enclosed. The equipment cabinet shall be built in such a way that it prevents the entry of all type of rodents.

The Contractor is required to provide and install all additional necessary indoor and outdoor cable raceways. The cable raceways shall be in conformance with the following:

Signal cabling and power cabling shall require separate cable raceways. Signal and power cabling shall not share the same raceways and shall be installed as far apart as is practical. Adequate shielding shall be provided as required. Especially, all the F.O. patch chords from FODB to FOTE cabinet shall be enclosed with minimum of 1” flexible standard steel hose.

The Contractor shall be required to provide protection from lightning and transient voltages for all wideband communications equipment. Suitable surge voltage protection devices shall be installed in all panels. 36-72V DC supervision relay (For supervising DC-I and DC-II source) having potential free contacts and audible alarm should be made available in the rack.

8.0 Earthing:

For each facility, the Contractor is responsible for meeting the following station and equipment earthing requirements:

(1) All safety earthing shall be in full compliance with EMI/EMC requirements as per relevant international standards.

(2) Each cabinet (enclosure) or cabinet (enclosure) group shall include suitable signal ground and safety earth networks. The signal ground network shall terminate at a separate signal ground stud connection isolated from safety earth.

(3) Each earth/ground network shall utilize copper bus bars, copper braids and/or16 sq mm or bigger earth cable. All equipment earth/ground connections shall be made directly to the equipment chassis utilizing grounding lugs and secured metal-to-metal with star washers. Use of the enclosure frame, skin or chassis mounting hardware as part of the earthing /grounding networks, is not acceptable.

(4) The safety earth network shall be connected to "earth ground" at the safety earth stud. The earth stud connection shall be sized for an external earthing cable equipped with a 2/0 solid copper lug secured metal-to-metal with star washers. Primary AC feeds and distribution within enclosures requires earthing wire connection to the safety earth stud.

(5) The Contractor is responsible for providing all required earthing/grounding cable and installation. Cabinet (Enclosure)and equipment safety earthing shall be subject to the Employer's approval. Dedicated earthing for the offered MPLS -TP equipment shall be provided by means of a separate earth pit in compliance with the standards mentioned for earthing the equipment and the separate earth pit of MPLS-TP equipment shall be connected with the earth mat of the substation through earth flat riser and flexible tinned copper strip.

NETWORK MANAGEMENT SYSTEM (NMS):

The network management system (NMS) shall have facilities to supervise, monitor, control and configure eachequipment and the whole network. It shall have capabilities of fault, configuration, performance and security management.

It shall provide various graphical views to the network such as geographical overview, logical network structure, and hierarchical view. The network management system shall allow defining different user profiles.

A comprehensive alarm management shall show current alarms such that icons of the network elements change their colors according to the alarm level. The alarms shall be categorized as critical, major, minor alarms and warnings. A summary shall indicate the total number of alarms and warningsin the entire network. An alarm list shall list all alarms of the entire network according to the time of their occurrence. It shall be possible to filter alarms with various filter criteria. Operators shall be able to add comments to the alarms.

The NMS shall support end-to-end service provisioning for MPLS-TP as well as for TDM based services. For MPLS-TP based services, (all service types VPWS, Tree, VPLS) shall be supported by the end-to-end service provisioning

functionality. The NMS shall perform channel calculations based on various parameters such as available bandwidth, number of hops, through put, provisioned bandwidth/services and channel diversity requirements**.**

- The adequate information about proposed TMN system architecture implying all relevant detailed data about compliance with mentioned ITU-T recommendations as well as about technology-specific (SDH/Data) issues must be supplied.
- Detailed system and interface documentation, including specification of their capabilities as well as the state of compliance with ITU-T and ETSI standards is required.
- State of compliance with other related standards, in particular with ISO,IETF and TMF, is required. The list of complied industrial standards must be provided.
- TMN system must include following levels and components concerning network management and operation:
- Element Management System(EMS),
- Network Management System(NMS),
- Remote Management Client (RMC),
- Local Craft Terminal (LCT)
- The EMS should represent the HW and SW components used for managing of one or more NEs.The EMS should provide management across a subset of NEs on an individual or collective basis, typically across single vendor equipment.
- The EMS must perform management functions across the Element Management Layer (EML, as per ITU-TM.3010) of the TMNLLA. The NMS should represent HW and SW components used for managing of an SMSN (SDH Management Sub-Network) as a whole.
- The NMS should provide end-to-end network view of the entire management domain enabling management of the NEs contained in the SMSN.
- The NMS must perform management functions across the NML of the TMN LLA. The NMS must have a Client/Server architecture enabling a scalable expansion of the Management capacity and tasks distribution.
- The Management System must have at least one NE that is connected to an OS, called a Gateway Network Element (GNE).An interface to establish this connection must be provided.
- The Management System must have Gateway Protection for GNE redundancy.
- Management solution must provide end-to-end view and manage of the entire domain.
- Both EMS and NMS must have MMI (GUI).Provided MMI must demonstrate similar look and feel for all SMSN involved. User-friendly MMI is essential.
- The offered management system must include a single unified NMS (network management system) and multiple EMSs (element management systems) for different product lines and/or scalability purposes of same vendors.
- Single EMS and NMS to manage the network on the Access Metro and core
- All data/SDH/Optical trails as well as maintenance actions should be managed by the NMS (top to bottom).

- The NMS must support Control Map to enable easy viewing and impact assessment of the various management channels and links carrying them. This includes DCC and GCC.
- The NMS must support upto 100NEs
- The NMS must support multiple physical and logical graphical views of the same network (Physical, Optical, ETH/MPLS, etc.) with Logical Element.
- The NMS must support Resource Domain Partitioning with fine granularity to allow different users access to different network resources (per NE/card/port/HO trail).
- The NMS must support Graphical network usage chart – graphically show used/free resources over the whole network or drill-down into a link. For links, it should be possible to display availability in case of network growth.
- The NMS must support Automatic PM History collection and built-in report generation (graphical and tabular) for Historical PM.
- The NMS must support multi-route trails (protection/restoration).
- The NMS must implement the Network Management Layer (NML) as defined in the TMN Logical Layer Model.
- The NMS must have a rich Graphical User Interface (GUI).It is required that it will be possible to perform any Network Management operation via GUI without the need for the user to perform Command Line Interface (CLI) commands.
- The NMS architecture must be based on client-server relationships using standard technologies.
- The NMS must scale to manage upto atleast 300 devices. List and describe the device management capabilities. Standalone and integrated Tele protection shall be manageable through same NMS.
- The management system should support display of:
- Data logical layer showing Switches and the topology
- The NMS will allow the provisioning, monitoring, operation and troubleshooting of the entire network from a single interface location. The single unified NMS must support management of MPLS, Describe this facility, addressing amongst others the ISO conceptual model of all key functional areas (fault management, configuration management,performance management, and security management) of network management.
- It should be possible to present 3rd party equipment as topology links and trails
- An upper level management system is required. Management functions must be based on TMN concepts of management layering (as perM.3010).This system will provide unified management capabilities to the entire managed domain. The interface to upper Management level should be based on the CORBA Technology or SNMP.
- Connectivity of the various management layers (NMS system and lower management systems, EMSs) must rely on a standard Management Communication Network (MCN).Both LAN and WAN connectivity must be enabled. Lower level management systems must be accessed directly or by MCN.
- The NMS should enable integration of additional equipment types (related technol-

ogies as well as multi-vendor sub-networks as per SNMP).

- Interfaces between EMS and NMS should be based on the CORBA technology and should relay on TMF guidelines (TM 509, TMF 513, TMF 608, TMF 814) that provide CORBA object models or SNMP.
- Insertion and Removal of NEs should automatic once the function activated
- The management system will have the ability to integrate 3rd-party SNMP equipment.
- The Offered TMN system shall be capable of integration to other OEM's NMS/UNMS upwardly through SNMP-NBI. The interface shall accommodate all required functionalities to implement any open source software (OSS) such as
  - (i) Real time formalities & Synchronization
  - (ii) Providing of Circuits
  - (iii) Upload of N/W topology.
  - (iv) Monitoring of circuits for topology & circuit changes
  - (v) Alarm handling

The OEM NBI has to support the PDH, SDH and MPLS-TP related data and is not limited to the following.

NBI Support required for Network Inventory:

1. NE details
   a. Name
   b. Location
   c. IP Address
   d. Product name(NE Model)
   e. Version
2. Physical Ports details
   a. Name
   b. Layer rates
   c. Directionality(Unidirectional/Bidirectional)
3. Hardware Inventory details
   a. Name
   b. Expected Equipment type
   c. Installed Equipment type
   d. Part Number
   e. Serial Number
   f. Version
   g. Occupying Holder name
   h. Holder Type(Rack/Shelf/Sub-Shelf/Slot/Sub-Slot)
   i. Holder State(In Service/Out of Service)
4. Protection Groups(MSP/MSSP)details
   a. Protection Group name

   b. Protection Group Type(MSP 1+1,MSP 1:n,2F-BLSR,4F-BLSRetc)
   c. Layer rate
   d. Reversion Mode
   e. Protection Group Parameters(MSSP Ring Node ID, WTR Time etc)
   f. Worker Port(s)
   g. Protecting Port
5. Topological Links details
   a. Layer rate
   b. Directionality
   c. A End Port details
   d. Z End Port details
6. Cross-Connection(As individual cross-connects for each network element) details
   a. Rate
   b. Cross Connection Type(2way, 3way and 4way)
   c. Directionality (Unidirectional / Bidirectional)
   d. Active status
   e. Source & Destination CTPs for both working and protecting direction
7. Sub-Network Connection (As trails, as exposed from under lying NMS/EMS) details
   a. Name
   b. Layer rate
   c. SNC Type(2way,3wayand4way)
   d. Directionality(Unidirectional/Bidirectional)
   e. Source & Destination details
   f. Route details(List of all cross-connects in this SNC)
8. Support to expose Ethernet bridges called as Matrix Flow Domains (MFDs)
9. Support to expose Ethernet services called Flow Domain Fragments(FDFrs)
10. Support for extracting and configuring the maintenance operation such Local Loopback / Remote Loopback.
11. Support for Trail Remaining, facility to set the Circuit label against each mapping.

NBI Support required for Alarms data:

The under lying NBI of NMS/EMS or the NEs should support the following.

1. Alarm notifications
2. Alarm synchronization

NBI Support required for Performance data:

1. Require Performance data from NMS/EMS NBI or from the NE level.
2. The U-NMS has the facilities to support Text NBI, FTP or any other vendor

proprietary NBI interface(s).

3. Should support 15mins performance data KPIs
4. Should support 24hrs performance data
5. Should support current performance data
6. Support facility to Enable and Disable PM data for any Physical ports, time slots.

NBI support for Activation / Deactivation:

1. NBI should support to create / modify the cross connections in the ports involved in the service.
2. NBI should support to create/modify the user labels of NE, Port, Cross Connections and Physical Links (Topological Links)
3. NBI support for Ethernet service upgrade, down grade and end point shifting support for Services.

System HW Requirements

- System must be designed in a way that would ensure that no database information would be lost at shutdown procedure.
- Back up management Workstation should be provided and ensure full Management Redundancy. All information must be mirrored automatically between the management stations in order to prevent data loss in case of a failure of the working station.
- Switching from active to the stand-by station and vice versa must be automatic or available only for the authorized operator. This action must be without any effect on the traffic.
- The server/workstation and craft terminal shall have suitable processor(s) which shall be sufficient to meet all the functional requirement and expansion capabilities stipulated in this specification. Only reputed make like Dell, IBM, HP make shall be supplied.

System SW Requirements

- EMS/NMS Operating system (OS) must be based on a widespread commonly accepted latest version of OS,which is also independently commercially available. OS upgrading must not affect traffic, database, communication SW, EMS/NMS SW or other SW.
- EMS/NMS database must be a relational database and SQL supportis mandatory. The database must be independently available .It must be possible to duplicate the EMS/NMS database. The upgrade of the database SW must be carried out without any data corruption and traffic interruption.
- EMS/NMS SW upgrading must be carried out without affecting traffic, EMS/NMS DB, OS or other SW.

- Functional concept of the proposed system must comply with the FCAPS Management Functional Areas function sets requirements defined in the ITU-T M.3400.
- SW reset (including hot reset) must not affect traffic or element configuration. Present SW configuration must be stored in each NE. It must be possible to store more than one SW version in each NE.
- In the case of management connection loss, traffic must not be affected and the performance monitoring data must be stored into NEs. This data must be available to the operator view when connection with the management station is re-established.
- After a power break, NE must always return to its predefined settings in designated time that must not be longer than 30 minutes. That designated time must herein be stated.
- It must be possible to remotely download and activate SW to all NEs from the EMS/NMS. Both downloading and activating SW processes must not affect traffic. In case of the activation of system software that may require rebooting of MPLS-TP equipment, the equipment shall restore to functionality and traffic within 5 minutes. This is stipulated to minimize the outage of Communication Equipment.
- It must be possible to download data from EMS/NMS to all NEs such as NE configuration. It also must be possible to upload data from all NEs to EMS/NMS such as: performance and NE configuration. These processes must not affect traffic.
- It should be possible to select a number of trails from the trail list and print all their endpoints and resources, as well as selected attributes.
- A Timing Map window should enable the user to conveniently monitor the network's timing topology.
- The NMS should provide Customizable /Standardized Alarms color codes.
- SW Enhanced Features
- A thorough description is required regarding the support and implementation of the following features in the proposed Management System:
- Automatic Path-finding End-to-End for SDH/MPLS-TP/ETHERNET/Optical trail creation at the NMS level using configurable criteria.
- Shared ducts allowing the user to specify the shared resources the link/trail may use. In case two links/trails use the same resources, the management system should be able to find out that the same resources are used by multiple links/trails.
- The Bidder shall specify if the management system can be used to complete the topology and represent external, passive or unmanaged NE and to which extent.
- A Sophisticated Customer Network Management (CNM) providing the ability to partition the managed network’s resources (domains) amongst various users/clients and assigning them different rights and capabilities (profiles) should be available. This will enable to lease ports, cards, or entire NEs to CNM clients, and give them the means to monitor or even manage these resources.

Performance Monitoring Requirements

- The following features should be provided:
- Performance monitoring data collection ,processing ,storage
- Source for Performance Monitoring (PM)data: TDM/Ethernet traffic

- PM data presentation using tabular and graphical report
- PM database management:
- The NMS must provide information for all its managed elements via a single north bound interface.
- The NMS north bound interface must support at least one of the following:
- TM Forum's Multi Technology Network Management (MTNM) TMF-814 CORBA based standard or SNMP.
- SNMPV3.
- The SNMP north bound interface must support alarms management functionality.
- The NMS must have 1 no. North bound and 1 no. south bound interfaces at present and shall support at least two (2) concurrent northbound/southbound interfaces in future.
- Different performance at the interface, device, and protocol levels will be collected on a regular basis. The polling engine in the network management system will be utilized for data collection purposes. The network management systems shall be capable of collecting, storing, and presenting polled data.
- The NMS will show all current values of 15M and 24HPM counters forth at trail.
- The management system shall support display and collection of all PM data and counters supported by the NEs.
- Current PM data shall be visible from the management system on a per trail/service basis.
- Historical PM data shall be collectable to files for export/browsing.
- Historical PM data for ETH ports shall be collected and displayed as a report per port with graphical display of the information
- Historical PM reports need to support export to CSV
- It will be possible to collect historical PM data into a dedicated database from which reports may later be retrieved in different cross-sections including:
- In table form
- In different aggregation periods
- For a given date range
- Per trail/service
- The management system shall support display and collection of multi technologies of optics, SDH and Data.

Reports and analysis

- The system must support inventory reports with details on total number of cards and sub cards installed in the network, the utilization of cards and ports in percentage and inventory attributes per element
- The system should provide detailed and flexible reporting capabilities to answer the following information requests:
- Total no. of free services lots
- Total no. of free client ports
- Cards and ports installation and removal trend over time–detailed per location, card type, operation, date.
- Total no.of installed cards and transceivers–in their name,type,category,location, part number, SW/HW version.

- Cards configuration mismatch report– where card were wrongly configured and hence do not function properly
- The system should have the capability to generate new, customized reports per demand
- The system should be able to send information by mail upon a threshold cross in any inventory value
- The system should be able to export reports in CSV/HTML /PDF/ formats.

NMS Hardware capability.

The network management system (NMS) shall have facilities to supervise, monitor, control and configure each equipment and the whole network. It shall have capabilities of fault, configuration, performance and security management. It shall provide various graphical views to the network such as geographicalOverview, logical network structure, and hierarchical view. The network management system shall allow defining different user profiles.

A comprehensive alarm management shall show current alarms such, that icons of the network elements change their colors according to the alarm level. The alarms shall be categorized as critical, major, minor alarms and of the warnings and a summary shall indicate the total number of them in the entire network. An alarm list shall list all alarms of the entire network according to the time of their occurrence .It shall be possible to filter alarms with various filter criteria. Operators shall be able to add comments to the alarms.

The DCN (Data Communication Network) to access all Network Elements shall be based on TCP-IP. - The management system shall offer an SNMP interface for alarm integration into higher order networks.

a) Local User Terminal

It shall be possible to connect the craft terminal to any Network Element in the network using the TCP-IP protocol. The craft terminal shall support configuration, maintenance, and status information.

b) Alarms

Each module shall supervise its functions and shall have an alarm-indication LED on its front. All alarms shall be collected by the NMS. Each node shall be capable of collecting up to 8 external alarms.

c) Test Loops

The equipment shall provide means to loop signals on 64kBit/s level as well as on 2Mbit/s level. It shall indicate an alarm if a loop is activated.

d) Maintenance facilities

Preferably every Network Element shall have a built-in Signal Generator and Analyzer to analyze communication paths. It must be possible to connect the Generator and Analyzer to the communication channels and terminate the signal on other Network Elements. It shall be possible to configure circuits locally with the craft terminal and remotely from the NMS or the craft terminal. It shall be

possible to loop-back signals locally and remotely using the craft terminal or the NMS.

System Capacity (To be supported):

Equipment supplied for the system shall be sized and equipped with sufficient capacity to support BOQ and shall require to support the capacity as specified here under. The system supplied shall be sized (to be equipped as specified) to support full system expansion.

Data communications channelization required to support the NMS specified in Technical Specifications(TS)are not identified in the appendices.Therefore,theContractor is required to size and equip the system to include all channelization and channel cards required to support the NMS function.

NMS Architecture

a) The Network Management System (NMS) structure:

NMS should be structured according to the NMS Logical Layered Architecture established in section 5 of ITU-T Recommendation M.3010 and cover the Element & Network Management layers of this architecture.

The system should be designed using a client-server paradigm and use a SQL complaint Database to store alarm and configuration data.

Access to the system shall be possible for many users at the same time using NMS Client software. Client software shall be installed on the server machine itself and on separate client PC.

The NMS main server shall be provided at Vidyut Soudha, Vijayawada. Appropriate convertors/cables/interfaces shall be provided by the bidder to achieve the connectivity between the equipments and the NMS server.

b) Platform and Redundancy

The NMS server software should be based on a secure operating system with latest version of software preferably Linux/Unix. The NMS software with life time free license for managing total nodes of about 100 nos. MPLS-TP. Any up- gradation in the software shall be done at free of cost. Patches/Service Pack up- gradation will be provided free of Cost as when required in life cycle of offered NMS Software. The NMS server hardware should have dual hard disks configured as a RAID system level 5 (mirroring).

There should be a possibility to add a second NMS server to create a Main/Standby system using an interconnecting link with a minimum bandwidth of 2Mbit/s. The Standby should automatically switch to active mode if it loses contact to the Main. It shall also be possible to operate the two NMS independently if the connection between the two servers is down.

For quick fault recovery and minimum NMS downtime it shall be possible to manually create a backup of the database and store it on the local disk or on a DVDs. A facility shouldexist to create a bootable backup of the complete system either on tape or DVDs.

c) Supported Network Elements

For ease of management and monitoring by APTRANSCO, the System shall be able to manage all the offered communication equipment and Tele-protection equipment from a single software and hardware.

d) Graphical User Interface

The Graphical User Interface should have definable network displays (maps). Icons symbolizing sub-networks, equipment and links are to be placed on an optional background image and can be opened to display greater details down to equipment port level. Context sensitive menus and online help should be available. It should be possible to create any number of network maps to create different views of the network, e.g. geographical, schematic, sub-network, etc.

The Bidder shall submit for APTRANSCOs approval the NMS architecture describing in detail the following sub systems/features**:**

i. Database used in NMS
ii. Master Processor, server/workstation, LAN, Peripherals and hardware
iii. Software and operating system
iv. Local Consoles/remote consoles
v. Craft Terminals
vi. Data communication between NEs, Remote /Local Consoles and TMN Processor(s)
vii. Routers /Bridges
viii. Expansion Capabilities

Management Functions

The NMS shall support following Management functions:

Configuration Management

Configuration management is concerned with management,display,and control of the network configuration. Minimum specific requirements that shall be satisfied include the following:

a. Provide tools to establish and maintain the backbone topology and configuration information and provide graphical maps depicting the configurations.
b. Gather descriptive information about the current configuration of the equipment, provide operator displays, and prepare reports.
c. Provide tools for planning, establishing, and changing the static equipment configuration. Provide for changes of the equipment configuration in response to equipment failures, planned upgrades, and operator requests to take equipment offline for testing.
d. Provide verification testing to support new equipment installation.
e. The through put current value, trends graph of each port for a period of minimum one year like daily, monthly and annually should be made available.

NMS Configuration

A graphical User Interface should exist to create network Domains, the NEs to be managed by the system and NE supervision processes. (Domains are used to structure the network in areas of responsibilities and are used in the definition of access right. The supervision processes allow grouping the NEs for supervision and enabling the supervision of a group of NEs to be turned to ON or OFF.)

NE Configuration

Once an NE is discovered by the NMS its configuration shall be automatically uploaded and stored on the NMS hard disk. It shall be possible to configure the NMS to automatically keep a definable number of the most recent NE configurations accessible for inspection and/or download. All NE configuration changes should be done from the NMS using easy to use graphical windows accessible via drop down menus starting from the NE symbol on the graphical network view. The NMS shall fully support the administration of the NE configuration data (upload, download, backup, copy) and the NE embedded software (upgrade).

End to End Circuit/Trail Configuration

As an NMS option a facility for automated end-to-end configuration of trails and circuits (trail management) shall be offered including the 64kBit/s cross- connections.

The option shall enable the operator to create and maintain database tables containing all the communication services configured into the network. A graphical user interface shall be available to browse, modify or delete the existing end to end connections or to create new ones .To create new connections it shall be sufficient to specify the type of connection, the required bandwidth, the protection type and the two end points. The software should then suggest routes through the network and can be used to automatically download the required cross-connections to all the NEs which belong to the selected route.

Trails e.g. at VC4, VC12 or E1 level can be inspected for the number of used/free channels.

Various reports shall be printed and information extracted in a format which is suitable to be imported into a spreadsheet program such as Excel.

Fault Management

The Network Management System should provide comprehensive and efficient fault management, based on a graphical network view and coloured symbols representing the alarm state of the respective network element.

Fault management is concerned with detecting, diagnosing, bypassing, directing service restoration, and reporting on all the backbone network equipment, systems, and links. Minimum specific requirements that shall be satisfied include the following:

a. Display equipment status in a consistent fashion regardless of the source of the data on a graphical topological, map-type display. Status shall be displayed through the use of colours on links and nodes as well as through text.

b. Obtain status and detect faults through periodic polling, processing of unsolicited alarms and error events, and periodic testing for connectivity.
c. Maintain an alarm summary of unacknowledged alarm events on the management station display and maintain a log of all received alarms. The operator shall be able to acknowledge and clear alarms individually and as a group. The use of alarm correlation techniques is encouraged to minimize the proliferation of alarms caused by a single, common event. All alarms shall be configurable as critical alarms, major alarms and minor alarms with different colours.

Additionally the following fault management features are required:

- The NEW symbols on the layered network views must have markers to indicate
  a) If an NE is reachable,
  b) Disabled through the operator,
  c) Has an attached operator note,
  d) if there are unacknowledged active alarms (one marker for each severity level),
  e) if there are unacknowledged cleared alarms
- An Alarm Summary View to display alarm counters per severity
- An Alarm List containing all active alarms and all cleared and unacknowledged alarms. The alarm list should be displayed in a tabular form with extensive filtering possibilities. It must be possible to adjust the size and the order of the displayed columns. Alarms are selectable for operator actions (e.g. acknowledgement) individually or in groups using the mouse and the Ctrl or Shift key.
- New active alarms should appear in the alarm list marked as unacknowledged until they are confirmed by the operator. New cleared alarms must be either marked as unacknowledged or alternatively self-acknowledged by the NMS software. The preferred behavior should be an NMS configuration option.
- For each alarm a detailed alarm view must exist to display all alarm properties including raise and clear times, the time of acknowledgement and the username of the person which performed the acknowledgment. It must be possible to add operator notes for each alarm.
- For external equipment with alarm output contacts (common alarm outputs, door contacts etc.) the fault management should provide an option to display the respective alarm states by colored symbols on the map.
- Sections (Links - typically fibre optic inter connection between nodes) should be dynamic and displayed in the colour of the most severe alarm of the ports which the link is connected to.

Synchronization Map

Since this is a large network, to facilitate the identification of synchronization problems within the communication network, the NMS should provide a dynamic synchronization

view/map. This synchronization view must display the active synchronization sources for every network element using a graphical representation, so the clock distribution in the network can be easily tracked.

Security Management

Access to the different functional areas of the NMS shall be strictly controlled by the privileges assigned to the individual NMS users. Accounts which shall have access to the NMS must be assigned NMS access rights by an NMS administrator. There shall be one overall NMS administrator which is able to create additional regional administrators which have full control e.g. to assign NMS access rights, but are limited to their domain – corresponding to e.g. a geographical region.

To simplify the task of access right assignments, customer specific user profiles can be predefined which then can be assigned to the individual NMS users. It should allow to create users which have access rights to several/all domains with a different profile for each domain.

Specifically, the following features must be individually assignable to different user classes and/or user accounts:

- Administrative functions such as adding, modifying and deleting NMS user accounts
- Different levels of access rights to the network elements (modify, monitor only, software download)
- Access to individual network elements or groups of network elements (regions)
- The right to create and modify network maps
- Use function such as Alarm Forwarding, Trail Configuration, etc.

Performance Management

Performance management is concerned with evaluation of the use of network equipments and their capability to meet performance objectives. Minimum specific requirements that shall be satisfied include the following:

a. Provide support for an operator to initiate, collect, and terminate performance metrics under both normal and degraded conditions. For example, BER of each link, together with other data measured at each node, shall be available on operator request.
b. Monitor point to point & end to end signal quality and history. Provide operator controls to monitor performance of specified events, measures, and resources.

Specifically provide displays to permit the operator to:

1. Select/deselect network equipments, events, and threshold parameters to monitor
2. Set monitoring start time and duration or end time
3. Set monitoring sampling frequency
4. Set/change threshold values on selected performance parameters
5. Generate alarm events when thresholds are exceeded.
6. Set multiple thresholds on certain performance parameters. Alarm categories include as a minimum a warning and a failure.

7. Calculate selected statistical data to measure performance on selected equipment based on both current and historical performance data maintained in performance logs. Performance data provided is limited to what is available from the equipment Contractors.
8. Provide graphical displays of point to point and end to end current performance parameter values. Provide tabular displays of current, peak, and average values for performance parameters.
9. Generate reports on a daily, weekly, monthly, and yearly basis containing system statistics in table and graphical form.

15 Minute and 24 hour Performance counters according to G.826 must be collected and stored on the NEs for a limited time. As an NMS option a function shall be available to cyclically collect these counters from the NEs and store them on the hard disk in XML format. The operator shall be able to specifically define which individual data collection points on which equipment shall be included in the collected datasets.

Interfaces to external systems

As a future option, to provide the possibility to send data to other NMS or to support additional non native equipment the system should support two open software interfaces, one to export data to external systems (northbound interface) and a second one to be able to feed basic equipment and alarm data to the NMS from external customer specific software (south bound interface).

Communication Channel Requirement and Integration

The Bidder shall provide Communication requirements for NMS system as a part of NMS system. The Bidder shall provide all required interface cards / devices, LAN, routers/bridges, channel routing, cabling, wiring etc. and interfacing required for full NMS data transport.

The NMS data transport shall utilize the wideband communications transmission system service channel in the overhead whenever possible. This will provide inherent critical path protection.

Should the configuration requirements dictate multiple NMS station processors, the NMS Master Station shall require bidirectional data transport with its station processor(s).This communications interfacing shall be via critically protected data channels. It shall be the Bidder's responsibility to provide for and equip all necessary critically protected NMS data channel support.

In case supervisory channels are not available, the Bidder shall provide suitable interfaces in their supplied equipment for transport of NMS data. The Bidder shallalso be responsible for providing suitable channels with appropriate interfaces to transport the NMS data.

The bidders shall describe in the proposal the NMS data transport proposed to be used by the bidder in detail including capacity requirements and various components /equipment proposed to be used.

Craft Terminal

Each MPLS-TP FOTE on the fibre optic communication network shall include provision for connecting a portable laptop to be known as craft terminal to support local commissioning and maintenance activities. Through the use of this laptops and local displays /controls, the operator shall be able to:

a. Change the configuration of the station & the connected NEs.
b. Perform tests
c. Get detailed fault information

The craft terminal shall be connected to the interface available in the communication equipment**.**

Hardware Requirements

Master Processor, Server/Work station and Craft Terminal

The server/workstation and craft terminal shall have suitable processor(s) which shall be sufficient to meet all the functional requirement and expansion capabilities stipulated in this specification. Only industrial grade reputed make Like DELL,IBM, HP make shall be supplied.

a.) The NMS & EMS on independent 1+1 redundant rack servers and each server shall have the following minimum configuration. Intel® Xeon® E-2388G 3.2GHz, 16M Cache, 8C/16T, Turbo (95W), 3200 MT/s Processor. 1x32GB UDIMM, 3200MT/s, ECC RAM. 2x800GB SSD SAS ISE Mix Use 12Gbps 512e 2.5in Hot-plug AG Drive, 3 DWPD.C3, RAID1 for 2 SSDs. Motherboard with Broadcom/Intel 5720 Dual Port 1 Gb On-Board. Dual, Hot-Plug, Redundant Power Supply (1+1), 600W. Red Hat Enterprise Linux Operating system with required permanent License & subscription. 19”LED Monitor However, the internal hard disk drive for the server shall be redundant and all the data shall be mirrored. Further, the NMS software shall support data mirroring on redundant disk drives.

b.) The workstation shall have minimum configuration of Intel Core i7-12700 processor, 16GB-DDR4-3200 MHz RAM, 256GB M.2 SSD and 1TB HDD @ 7200 rpm, 1G Ethernet port, serial port, USB Port, HDMI port, A/V port, USB-Wired keyboard and mouse, 21.5” LED monitor IPS. Appropriate network drive card shall also be provided wherever required.

CPU enclosures shall be desktop type and shall include available expansion slots except for the Craft Terminal which shall be a laptop.

c.)The craft terminal shall have minimum configuration of Intel Core i7-12700 (4.7 GHz, 24 MB Cache), 16 GB DDR4 RAM, 500GB SSD, 4GB graphics, DVD-RW drive, built in webcam, built in speakers, Keyboard, mouse, USB Port-2 No., HDMI port, A/V port, 1G Ethernet port, 15” monitor and battery back-up of at least 4 hours. Appropriate network drive card shall also be provided wherever required**.**

Peripherals and hardware

NMS system consists of 1 No. main server, redundant server, work station, craft terminal to be installed at Vidyut Soudha, Vijayawada).

Local/Remote Operator Consoles

The Bidder shall provide operator consoles sized and equipped to support the system furnished and incompliance with the specification. The console shall Provide hardware interfacing for the NMS users to the software operating support systems. At a minimum, a console shall include the hardware similar to a work station.

Power Supplies

The NMS system shall use 220 volts 50 Hz A.C or 48 volt D.C as available at site for its operation as available at site.

General Software/Firmware Requirements

Due to various alternative design approaches, it is neither intended nor possible to specify all software and firmware characteristics. It is the intent herein to provide design boundaries and guidelines that help to ensure a demonstrated, integrated program package that is maintainable and meets both hardware systems requirements and the customer's operational requirements.

Operating System Software

Operating system software shall be provided to control the execution of system programs, application programs, management devices, to allocate system resources, and manage communications among the system processors. The BIDDER shall make no modifications to the OEM's operating system, except as provided as USER installation parameters.

Applications Software

All application essential software for proper functioning shall be written in a high-level programming language unless developed using industry proven application programs and development tools provided with the system. The each craft terminal(laptop)shall be provided with MS office 2016 basic OEM single and Antivirus .The BIDDER shall make no modifications to the applications program except as provided as USER development tools.

Software Utilities

A utility shall be provided to convert all reports into standard PC application formats such as excel.

Revisions, Upgrades, Maintainability

All firmware and software delivered under this specification shall be the latest field proven version available at the time of contract approval. Installed demonstration for acceptance shall be required. All firmware provided shall support its fully equipped intended functional requirements without additional rewrite or programming.

All software shall be easily user expandable to accommodate the anticipated system growth, as defined in this specification. Reassembly, recompilation or revision upgrades of the software or components of the software shall not be necessary to accommodate full system expansion. All Software provided shall be licensed software and shall be compliant with national and international industry standards.

Database(s)

The contractor shall develop all the databases for the final wideband network following the global acronyms for all stations. Database(s) to be provided shall contain all structure definitions and data for the integrated functional requirements of the NMS system.

NMS operator Groups shall share the same virtual database. This means that they shall share the same database and database manager, whether or not physically separate databases are maintained.

HELP:

All applications shall be supported by USER accessible HELP commands that shall assist the user in the performance of its tasks. HELP commands for an application shall be available to the user from within the active application and shall not interfere with the activities of the application.

SYSTEM MAINTAINABILITY

To facilitate performance trending, efficient diagnosis and corrective resolution, the system shall permit in-service diagnostic testing to be executed both locally and from remote locations, manually and/or initiated under NMS control. Such testing shall not affect the functional operation of the system.

Preventive and problem oriented maintenance of the communications system shall be performed using diagnostics tools such as NMS and test equipment (Optical power meter, communication analyser etc.). They shall support complete maintenance of all system elements and shall permit the diagnosis of any without requiring additional test equipment. The Contractor shall provide specialized training required to operate above mentioned diagnostic tools. For all redundant systems, disconnection and repair of any failed device shall not interrupt the operation of the system.

SYSTEM UPGRADABILITY AND EXPANDABILITY

Equipment supplied shall be sized for at least a backplane capacity of 40G/100G Power supplies and NMS shall be sized for maximum equipped system capacity.

11.0 EQUIPMENT LIFE SPAN:

All equipments supplied shall have a minimum expected life of fifteen (15) years from the date of operational acceptance.

12.0 REVISION LEVELS AND MODIFICATIONS

All hardware, firmware and software delivered as part of the communications network shall be field proven and at the most current revision level. All modifications and changes necessary to meet this requirement shall be completed prior to the start of the factory tests or under special circumstances, on written approval of APTRANSCO, prior to the completion of SAT.

All field modifications of the hardware, firmware and software that are required to meet installation and/or performance specifications, shall be fully documented as part of the deliverables, both as a separate field modifications record and as corrected equipment/ configuration documentation. All the updates of software related to this project then and

there should be provided at free of cost until the expiry of life span of the equipment /NMS supplied under this project.

13.0 FODP to MPLS_TP Equipment/ Optical Amplifier Connectivity

The Contractor shall be responsible for connectivity between the FODB and the FOTE equipment. The Contractor shall provide FC/BC coupled patch cords. The location of FODB shall be finalized during detailed engineering.

The patch-cord length between the FODB & equipment rack shall be suitably protected from rodents, abrasion, crush or mechanical damage .Preferably steel flexible hose shall be provided as per requirement.

14.0 CES (Circuit Emulation Service) Support:(Where ever applicable)

The equipment shall fully comply with SAT oP (RFC 4553) and CES oP SN (RFC 5086) and support the following CES (Circuit Emulation Service) functions over ETH and MPLS-TP:

- The E1CES card shall have at least 8 ports.

- On TRANSPORT LEVEL the equipment shall support / future upgradable to the following connection ports:
  - Upto 4x1Gbit/s MPLS-TP capable transport ports
  - Nx2Mbit/s electrical E1ports, G.703/IECC.37.94for differential tele protection.
  - 8 port Ethernet10/100/1000Mbps.
  - All optical MPLS-TP and Ethernet ports shall support SFP or SFP+ modules (small-factor pluggable unit) for short, medium, long and extra-long optical communication. ALS (Automatic Laser Shut-down) shall be fully supported.

15.IP/Ethernet Interface

Ethernet module shall comply with the following specification: Ethernet electrical connection: 10/100BaseTEthernet optical connection: 1000Base-LX/SX

Switching: by pass mode for IEEEStd802.3 frame or based on port or VLAN tagID

This module shall provide WAN connectivity of VC-12 or VC-3 and also support configuration of Logical WAN port.

Framing: According General Framing Procedure (GFP) ITU-TG.7041 Capacity: Virtual Concatenation (VCAT) acc. ITU-TG.707

Protection: Link Capacity Adjustment Scheme (LCAS) acc.ITU-TG.7042.

Alarm Interface.

The module shall provide means to collect at least 8 alarms, which shall be displayed on the Network Management System. It shall be used to supervise external equipment by the Network Management System. A minimum of 2 outputs, which can be controlled from the Network Management System, shall be available.

It shall be possible to label an alarm. The label-text shall be read from the interface module and visible on Network Management System as well as on the local craft terminal. The equipment failure shall be indicated with one audible alarm.

13.0 MDF, DDF AND CABLING

For the purposes of the specification, the contractor shall provide cabling, wiring, DDF patching facilities and MDFs interfacing to the wideband telecommunications system. Equipment and material components for MDF, DDF and cabling are also part of this procurement. It shall be the Contractor's responsibility to provide all cable support required for full supplied equipment interconnection with the MDF and shall be in accordance with communications industry standard practices and the requirements mentioned in the technical specifications.

14.0 MDF AND DDF PATCHING FACILITIES

The Contractor shall supply and install all cabling, wiring, connectors, cross connects, Digital Distribution Frames (DDF) and Main Distribution Frames (MDF) associated with the installation and interconnection of equipments procured under this package.

a. Any other cables, connections etc required for a fully functional, integrated telecom system and spares as per site requirement.

The connections amongst various equipment such as FOTE and subscriber MDFs etc. shall always be routed through DDF and MDF to provide maintenance access.

Management Interfaces

The equipment shall make provision for an Ethernet port (hence forth called “local management interface”) that allows access only to the local management agent (i.e. to the management agent running on the equipment itself but not to management agents running on remote equipment).

The equipment shall make provision for interface(s) that allow access to the management agent from a remote site through an IP network (henceforth called “routed management interface(s)”).

The equipment shall make provision for a VLAN dedicated exclusively to the traffic to /from the management agent (out-of-band management). The VLAN identification must be configurable.

The priority of the traffic flowing through the routed management interfaces shall be configurable (i.e. capability to assign a Class of Service to the management traffic).It shall be possible to enable/disable the local management interface by configuration. It shall be possible to enable/disable the routed management interface(s) by configuration. The IP address of the local management interface shall be configurable with proper authentication. The IP address/addresses of the routed management interface(s) shall be configurable. The local management interface shall be browser based one. It shall not participate in dynamic routing protocol (i.e. the local interface and its IP address are not propagated outside the local equipment).It shall be possible for a routed management interface to participate in a dynamic routing protocol. The participation shall be configurable.

The equipment shall support the following protocols for the communication between the management agent and the network management system (NMS)/craft terminal (CT):

a. Telnet

b. SSH

c. SNMPv3

d. SYSLOG

The equipment shall support the capability of sending event messages to up to ten different remote syslog servers. Each server is identified by its IPv4 address.

OSPF authentication

The equipment shall provide the capability to authenticate OSPF messages (e.g. MD5).This capability shall be provided for both DCN and user traffic OSPF capable routers. The DCN protocol of all OEMs which are designed as per relevant IEC/IETF/ITU-T standard and satisfy the requirement of APTRANSCO will be accepted.

e. Cyber Security Compliance:

The fibre optic terminal equipments proposed in this tender shall comply all the cyber security requirements defined in the IEC standard 62443-3-3.Necessarytype test report obtained from Indian Govt. laboratory (CPRI)/ Authorized International laboratories for the above cyber security features shall be uploaded with the bid.

1. Other mandatory test:

As per the order dated 02.07.2020 of the ministry of power, Govt of India that apply to any item imported for end user or to be use data component or as a part in manufacturing, assembling of any equipment or to be used in power supply system or any other activity directly or indirectly related to power system the compliance to following requirements are to be ensured. In this context all expenses towards the mandatory testing is to be borne by the project contractor.

a. All equipment, components and parts imported for use in the power supply system and network shall be tested in India to check for any kind of embedded malware/Trojans/cyber threat and for adherence to Indian standards.

b. All such tests shall be done in certified laboratories that will be approved by the ministry of power, Govt of India.

c. Any import /equipment/ components/ parts from:“Prior reference” countries as specified or by persons owned by controlled by or subject to the jurisdiction or the direction of these “Prior reference” countries will require prior permission of the Govt of India.

d. Where the equipment /components /parts are imported from “Prior reference” countries with special permission , the protocol for testing in certified and designated laboratories shall be approved by the by ministry of power, Govt of India.

It is mandatory for the bidders /OEM from prior reference countries to obtain the type test report for having tested for cyber security features as per relevant standard from CPRI laboratory of India at their own expenses. As per CEA guidelines 2021,

18.0Equipment Availability

The availability requirements are categorized in two parts. The Contractor shall ensure the availability for subscriber to subscriber where they will have to provide the new equipment (MUX/Drop-Insert) under this contract. The availability requirements areas follows, which shall be demonstrated at site for the equipment being provided under this contract:

A. New nodes with requirement of MPLS-TP equipment:-

(1) The average per link subscriber to subscriber availability shall be atleast 99.97%.The average per link subscriber to subscriber availability is defined as the availability between any two data or voice subscribers/Tele-protection channel.

(2) The network-wide subscriber to subscriber availability shall be at least 99.8%.The network-wide subscriber to subscriber availability is defined as the availability between any two data or voice subscribers /Tele-protection channel on the wideband network.

The calculated availability is defined as the theoretical availability determined by a statistical calculation based on the mean-time-between-failure (MTBF) and the mean- time-to-repair (MTTR) of the components and modules comprising the FOTE. The down time of the fibre optic cable shall not be considered in the aforesaid availability calculations.

In order to ensure that the equipment & configuration proposed by the bidders shall be capable of demonstrating the specified availability figures, it is required that the Bidders shall include in their proposal a calculated availability analysis for the proposed equipment. The calculated failure rates of the units and the calculated availabilities of the equipment being offered shall be provided in the proposal. The analysis shall be based on an availability block diagram and shall include the mean-time-between failure(MTBF) and mean-time-to-repair (MTTR) of all of the components on the link. The Contractor shall indicate in the analysis the MTBF and MTTR and the resulting availability of each point-to-point link. For this analysis, an MTTR of at least 4 hours shall be assumed.

19.0 Lost Signal Recovery

At any digital signal level, reapplication of a lost signal shall result in automatic resynchronization and full restoration to normal operation without manual intervention. All alarms incident to the signal failure, shall be automatically cleared at the equipment, rack and monitoring levels and normal operation indications restored and reported if applicable.

20.0 TESTS

TYPETEST

- The MPLS-TP FOTE offered shall be fully type tested within five years in any Indian or International accredited laboratory as per the relevant standards. The bidder shall submit the complete type test reports for the offered MPLS-TP FOTE along with the offer **otherwise the offer shall be liable for rejection.**

- The purchaser reserves the right to demand repetition of some or all the type test in the presence of the purchaser's representative. For this purpose the bidder may quote unit rates for carrying out each type test.

21.0 DOCUMENTATION for NMS:

The bidder shall have to submit the working diagram (hard copy as well as Soft copy) for the NMS system comprising various nodes of the OPGW network showing a connection diagram with details of interface port from central control stations to nodes of the network. (2 Copies)

Drawings shall be of A-3 size. (297mmX420mm). Final Drawing, Technical literature and commissioning manuals shall be printed and not photocopy. Each consignment shall be accompanied by a detailed packing list containing the following information.

- Purchase order reference.
- Name of consignee
- Details of consignment
- Destination
- Total weight of consignment
- Handling and unpacking instructions.
- Bill of materials indicating contents of each package
- Sign showing upper /lower side of the crate.

For any deviation from the specification which is not specifically brought out in the schedules shall be supported by authentic documents, standards and clarifications; otherwise the offer may be liable for rejection.

22.0 Test Documentation

The Contractor shall provide documentation for all factory and field tests. The test documentation shall include the following:

a. Test Procedure Document
b. Type test documents
c. Factory Acceptance Test Documents
d. Site Acceptance Test Documents
e. Availability Test Documents

Factory Acceptance Test results will be included in factory test data. Test results, for any un-witnessed equipment tests, shall be shipped along with the associated equipment. All FAT and SAT test documents shall be got approved before the dispatch of first lot of equipment.

23.0 SPARES

The successful bidder shall furnish unit rates for mandatory spares and also for various types of all modules, SFP+, Boosters, Wave length translator etc. along with the price bid after award of contract .Also bidder shall have to submit the offer for the spares other than specified in BOQ if required, with unit price separately. The bidder is bound to supply for maintenance as and when required for service life of the MPLS-TP FOTE & NMS.

Note: All SFPs shall be of MSA2000 compliant in order to work with any vendor FOTE equipment.

24.0 Summary of applicable Standards:

The equipment shall conform to the following standards

1. Immunity

- IEC/EN61000-4-2:Electro static discharge Immunity, level 6/8KV
- IEC/EN61000-4-3:Immunity to electromagnetic field radiated ,level 10V/m
- IEC/EN61000-4-4: Immunity against electrical FastTransients,Level4KVon DC i/p and all other ports.
- IEC/EN61000-4-5Immunity against voltage surges, DCterminals.Level 2KV,others 1KV.
- IEC/EN61000-4-6 Immunity against radio frequency voltage conducted in Power terminals, level 10V/m
- IEC/EN61000-4-17 ripple on DC input port Immunity
- IEC/EN61000-4-18 Damped Oscillatory wave Immunity
- IEC/EN61000-4-29 Immunity against short voltage dips, interruption and voltage variations

2. EMISSION:

- EN55022(CISPR-22)conducted emission, Level Class B
- EN55022(CISPR-22)radiated emission, Level Class B

3. SAFETY

- IEC/EN60950-1Information technology equipment safety.
- IEC60255-27Measuringrelaysandprotectionequipmentproductsafetyrequirement.

4. MECHANICALTEST-TRANSPORT (IEC60721-3-2CLASS2M1)

a) Vibration 5-9 Hz displacement amplitude IEC 60068-2-6 +/-0.3mm
(Sinusoidal) 9-200Hz displacement amplitude 1 m./sec sq.

b) Shock-Shock response spectrum type L IEC 60068-2-2740 m/ secsq
Peak acceleration

c) Freefall -Height of fall IEC60068-2-31 0.25m

5. ETS/EN

- EN60825-2:Laser Safety

6. ITU-T

- ITU-TG.664: Optical safety procedures and requirements for optical transport systems.

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GUARANTEED TECHNICAL PARTICULARS AND REQUIREMENTS–GTP FOR MPLS-TP FOTE EQUIPMENTS

GENERAL		REQUIRED	OFFERED
Type of multiplexer		L2/MPLS-TP, Legacy TDM	
Complying to IEC/ ITU-T/ IETF recommendations		YES	
Transmission Capacity	Gbit/s	1Gbit/s upgradable to 10Gbit/s MPLS-TP/L2	
Packet switching capacity	Gbit/s	Minimum 40Gbits/sec meeting the cross-connect capacity	
Hardware supported QoS scheduling		YES	
SNMP north bound interface supported		YES	
Integrated cyber security features		YES	
Equipment used in substation environment		YES	
The equipment is type tested by an independent, accredited test laboratory(e.g. VDE, DNV/KEMA)		YES	
Redundancy of protection signal transmission over MPLS-TP at port level		YES	
PACKET SWITCHING FEATURES			
Number of transit LSPs		≥128	
Number of terminating (protected) tunnels		≥8	
Number of VPWS instances		50	
Support of MPLS-TPP2MPtree service		NO	

Number of fast BFD sessions(≤10ms)		0	
Number of medium BFD sessions(≤50ms)		0	
Number of slow BFD sessions(≥100ms)		8	
1:1 bidirectional, symmetrical LSP protection		YES	
Support of L2GP protocol		YES	
L2GP dual homing support		YES	
Support of RSTP protocol		YES	
Number of supported VLANs	Amount	≥4000	
VLAN tag stacking support		YES	
MAC address table size	Amount	≥8000	
SYNCHRONIZATION			
Support of Synchronous Ethernet		YES	
Support of IEEE1588v2PTP ordinary clock		YES	
Support of IEEE1588v2PTPboundaryclock		YES	
Support of a ITU-T 8275.1/ 8275.2 telecom profile compatible PTP profiles		YES	
Accuracy of PTP time of day information per node	µs	≤1,5µs	
PACKET TRANSPORT INTERFACES			
Optical10Gbit/s MPLS-TP interface	Amount	≥2	
Optical1Gbit/s MPLS-TP interface	Amount	≥6	
Alarm Ports	Amount	Yes	

Circuit emulation module			
Complying to ITU-T G.703/ G.704, configurable		YES	
HDB3linecodesupport		YES	
CES PW IF available for serial and telephony access interfaces		YES	
SAT oP method support		YES	
POWERSUPPLY			
Operation	VDC	48V(-15%/+20%)	
1+1 Power Supply Redundancy/ Dual Power Feeding Option.		YES	
CONFIGURATIONTOOL			
Local/remote operation		YES/YES	
Data communication network(DCN)		Ethernet/IP	
NETWORKMANAGEMENTSYSTEM			
Fault management		YES	
Configuration management		YES	
Alarm management		YES	
Inventory management		YES	
Security management		YES	
Synchronization management		YES	
Ethernet service andQoS management		YES	
Data communication network(DCN)		Ethernet/IP	
NMS configurations, user interface configurations, low level and high level commands in main servers, works station, LCT etc., software configuration tools(both front end and back end), displays, management tools etc., are in English Language only.		YES	

NMS should support third party UNMS over NBI(North Bound Interface) for all management functionalities.		YES	
Inter operability with other MPLS-TP FOTE vendors is possible at all levels of band width. Necessary Proof of concept document is furnished.		YES	
Multi Source agreement (MSA) standard SFPs/SFP+/smart SFPs will be provided which is suitable for all MPLS-TP FOTE Equipment vendors.		YES	
The equipment shall restore to functionality and traffic within 5 minutes in case of		YES	
AMBIENTCONDITIONS			
Storage:ETS300019-1-1,class1.2	°C / %	-25…+70°C/95% ROH(non condensing)	
Transport:ETS300019-1-2,class2.2	°C / %	-25...+70°C/ 95% ROH(non condensing)	